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Wooden Wedge Model



A 3D model of a wedge shape made out of wood that can be used in various education settings. This wedge was used in a University-level Multivariable Calculus class to visualize setting up triple integrals.

Materials and Costs:

- **Wooden Dowel**: 2 in x 48 in, ~$11 USD at Lowe's/Home Depot.
- **Miter Saw** (or any other electric angle saw), ~$200 CAD.
- **Sand Paper** (various grits), ~$5 CAD.

Prototyping Process:

- It took several tries to create a wedge with the perfect lower "lip". We had difficulties where we made the lip too large, or cut it off completely. The trick was to set up a "stabilizing" structure around the chop saw where the large wooden dowel could lay on while we were cutting.
- We tested cutting the wedge at both a 45 degree and 30 degree angle. Although the 30 degree cut was more aesthetic and "cheese-like", the 45 degree cut was easier to mass-produce as one cut yielded two models.

Step by Step:

1. Measure and mark a length of 2 inches from the bottom of the dowel with a pencil (for a 45 degree cut with a 2 inch diameter dowel; adjust as desired).
2. Angle the chop saw. The angler can usually be found at the rear of the saw.
3. Place your wooden dowel on the saw, with the 2 inch mark face up and under the saw blade.
4. Create a sturdy surface for the wooden dowel to lay on. Because the dowel is 4 feet long, it cannot simply hang off of the side of the raised chop saw. Place other supporting wood under the dowel so it is level with the chop saw, and is sturdy and stable enough to be cut through, and will not move. Think of creating a "bed" for the part of the dowel that is not on the chop saw.
5. After your dowel is stable, make your cut at the mark. If your model has a lip that is too large or non-existent, decide where the next cut should be: perhaps you should cut below or above your mark instead. Use trial and error until you get consistent results and a "feel" of how your saw cuts, and how close to the mark you should be cutting.
6. Once you've made a 45 degree cut, angle your saw back to 90 degrees (completely upright). Move the dowel forward until you reach the un-angled part of the model, and cut straight across, eyeballing it to ensure the right sized lip.
7. Go back to step 1 and repeat until you've made as many as needed!
8. Use sandpaper to smooth the model's surface.

Tips and Tricks:

- If you consistently overestimate the lip and it is too big, use sandpaper to shorten the lip and even it out. We used an electric sand-paper tool to help us do this.
- Start out with large grit sandpaper, and then use finer grits until the model is smooth to the touch.
- The dowel we used was made out of poplar wood, which would occasionally chip when cut. At the hardware store, there were more options for a more expensive, higher quality wooden dowel.
- 80 models took about 5 hours to make (including prototyping, cutting, sanding and packaging).







